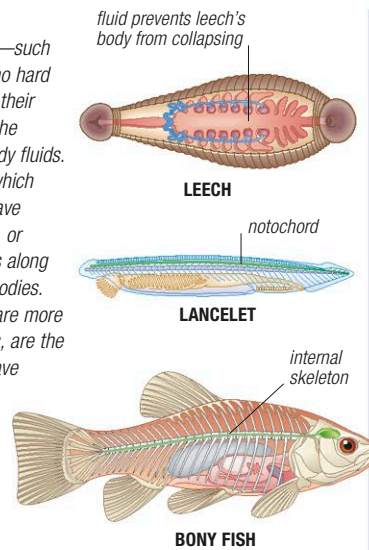


SUPPORT SYSTEMS

Many invertebrates—such as leeches—have no hard body parts; to keep their shape they rely on the pressure of their body fluids. Simple chordates, which include lancelets, have a strengthening rod, or notochord, that runs along the length of their bodies. Vertebrates, which are more advanced chordates, are the only animals that have internal skeletons made of bone, and/or cartilage.



are tiny, and many live in inaccessible habitats, which explains why they are still poorly known compared with chordates.

Warm- and cold-blooded animals

Most animals are cold-blooded (or ectothermic), which means that their body temperature is determined by that of their surroundings. Birds and mammals are warm-blooded (or endothermic), which means that they generate their own heat and maintain a constant internal temperature regardless of the conditions outside.

This difference in body temperature has some far-reaching effects on the way animals live, because animal bodies work best when they are warm. Cold-blooded animals, such as reptiles, amphibians, and insects, operate very effectively in warm conditions, but they slow down if the temperature drops. They can absorb some heat by basking in sunshine, but if the temperature falls below about 50°F (10°C) their

**HEAT CONTROL**

A basking butterfly soaks up the sunshine. By basking, or by hiding in the shade, butterflies and other cold-blooded animals can adjust their body temperature. Even so, they have difficulty coping with extreme temperatures, especially severe cold.

muscles work so slowly that they find it difficult to move. Birds and mammals, on the other hand, are barely affected by this kind of temperature change. Their internal heat and good insulation enable many to remain active even when temperatures fall below freezing.

Individuals and colonies

Physically, most chordates function as separate units, even though they may live together in families or larger groups. In the invertebrate world, it is not unusual for animals to be permanently linked together, forming clusters called colonies. Colonies often look and behave like single animals. Most are static, but some—particularly ones that live in the sea—can move about. Colonial species include some of the world's most remarkable invertebrates. Pyrosomes, for example, form colonies shaped like test tubes that are large enough for a diver to enter. But, in ecological

**LIVING TOGETHER**

This branching coral is covered with a living "skin" that connects its individual animals, or polyps. The polyps are in constant contact with each other but otherwise lead separate lives. Each has a set of stinging tentacles and catches its own food.

terms, the most important colonial animals are reef-building corals, which create complex structures that provide havens for a range of other animals. In reef-building corals, the members of each colony are usually identical. But in some colonial species, the members have different shapes designed for different tasks. For example, the Portuguese man-of-war—an oceanic drifter that has a highly potent sting—looks like a jellyfish but consists of separate animals, called polyps,

that capture food, digest it, or reproduce. They dangle beneath a giant, gas-filled polyp that acts as the colony's float.

Fuel for life

Animals obtain their energy from organic matter, or food. They break food up by digesting it through chemical reactions using oxygen, and then they absorb the substances that are released. These substances are carried into the animal's cells. This process—called cellular respiration—is like a highly controlled form of burning, with food acting as the fuel.

The majority of animals are either herbivores, which eat plants, or carnivores, which eat other animals. Carnivores include predators, which hunt and kill prey, and parasites, which feed in or on the living bodies of other animals. There are also omnivores, which eat both animal and plant food, and scavengers, which feed on dead matter—from decaying leaves and corpses to fur and even bones.

All animals, regardless of lifestyle, ultimately provide food for other animals. All are connected by food chains, which pass food—and its energy—from one species to another. However, individual food chains are rarely more than 5 or 6 links long. This is because up to 90 percent of an animal's energy cannot be passed on: it is used up in making the animal's own body work.

CARNIVORE 3

The food chain ends with a "top predator"—in this case, an osprey. When it dies, the energy in its body is used by scavengers, such as insects and bacteria.

**CARNIVORE 2**

The perch lives almost exclusively on other animals. A perch feeding on dragonfly nymphs is a second-level carnivore, receiving food that has already been through 2 other animals.

**CARNIVORE 1**

Dragonfly nymphs are typical first-level carnivores, using a mixture of speed and stealth to hunt small prey. Tadpoles are a good food source and often feature in their diet.

**HERBIVORE**

During their early lives, tadpoles use their jaws to feed on water plants. By digesting the plant food, and therefore turning it into animal tissue, they change plant food into a form that carnivores can use.

**PLANT**

By capturing the energy in sunlight, plants transform it into food to drive life on earth. In this chain, waterweed is the first link, creating food that can then be passed on.

**FOOD CHAIN**

This is a typical 5-stage food chain from a freshwater habitat. Food and energy move upward through the chain, being passed on every time one organism eats another. The chain ends when it reaches an animal that has no natural predators. The energy in the final animal's body is ultimately passed on to scavengers and to recyclers, known as decomposers, many of which live in soil.

INSULATION

In sub-zero conditions, superb insulation keeps the body temperature of these young emperor penguins at an almost constant 104°F (40°C).

